Atul Kumar

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Profile

Aspiring Firmware and Embedded Systems Engineer with hands-on experience building ESP32-based IoT systems, sensor integration, and real-time embedded firmware in C, complemented by Python and full-stack web development skills. Quick learner with strong problem-solving, communication, and teamwork abilities.

Education

Godavari Institute of Engineering and Technology *2023 – 2027*
B.Tech in Electronics and Communication Engineering (CGPA: 7.5)

Govt Higher Secondary School, Garh Dhamma *2021 – 2023*
12th (80%)

Experience

Intern – Embedded Systems / IoT, Genz Educate Wing *Feb 2026 – May 2026*

- Developed the Adaptive Smart Traffic Signal Controller project using ESP32 and FSM-based signal control logic.
- Worked on embedded systems, sensor integration, and real-time simulation using Wokwi Simulator.

Projects

Smart AI-Powered Traffic Signal Controller View Live → *Python, YOLOv8, OpenCV, Flask, ESP32, React*

- Built a real-time vehicle detection pipeline (YOLOv8, OpenCV) to estimate traffic density and adjust signal timing (10s–60s) dynamically.
- Integrated ESP32 with a Flask REST API to control physical traffic LEDs; built a React dashboard for live stats.

PlantCare AI – Plant Disease Monitoring System View Live → *Python, TensorFlow/Keras, OpenCV, Flask, Next.js*

- Trained a custom CNN to classify plant leaf diseases across 71 crop-disease categories, using a 2-stage hierarchical inference pipeline to eliminate cross-crop false positives.
- Ran inference on a local Flask API server, with results relayed over local IP to the Next.js frontend for instant on-screen diagnosis.

SensorNest – Smart Device Monitoring Platform View Live → *ESP32, C, FreeRTOS, I2C, Python, Next.js, MongoDB*

- Built a full-stack IoT platform: ESP32 firmware (C, FreeRTOS, I2C) reads DHT22/MPU6050 sensors, with OLED status display.
- Programmed a Python bridge to fetch live sensor data from the ESP32's local IP web server and relay it to a cloud-hosted Next.js + MongoDB backend, with authentication and TTL-indexed 7-day data retention.

Skills

- **Programming:** C, Python, JavaScript, C++ (Basics)
- **Embedded / IoT:** ESP32, ESP-IDF, FreeRTOS, I2C, Arduino IDE, Wokwi Simulator, Sensor Integration (DHT22, MPU6050, PIR, LDR)
- **Web / Full-Stack:** Next.js, React, Node.js, MongoDB, Flask
- **Tools:** Git (Basics), VS Code
- **Soft Skills:** Problem Solving, Communication, Teamwork, Adaptability
- **Languages:** English, Hindi, Telugu

Certifications & Achievements

- Blockchain and Applications, Prompt Engineering, Python Programming – Great Learning
- Cloud Fundamentals – Skill Camp
- Head Boy, Govt Higher Secondary School Garh Dhamma (2021–2023) — PMSSS Scholarship Recipient